

(1) Find a potential function for the vector field $F(x, y) = x^3y^4\mathbf{i} + x^4y^3\mathbf{j}$ and use it to evaluate the line integral of F along the curve $C(t) = \sqrt{t}\mathbf{i} + (1+t^3)\mathbf{j}$ for $t \in [0, 1]$.

(2) Find a potential function for the vector field $F(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + (xy + 2z)\mathbf{k}$ and use it to evaluate the line integral of F along the line segment from $(1, 0, -2)$ to $(4, 6, 3)$.

(3) Using Green's theorem, evaluate

$$\int_C (3y - e^{\sin x}) dx + (7x + \sqrt{y^4 + 1}) dy$$

where C is the circle $x^2 + y^2 = 1$.

(4) Using Green's theorem, evaluate

$$\int_C (y + e^{\sqrt{x}}) dx + (2x + \cos y^2) dy$$

where C is the positively oriented boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$.

(5) Find (a) the curl, and (b) the divergence of the following two vector fields:

- (i) $F(x, y, z) = xyz\mathbf{i} - x^2y\mathbf{k}$
- (ii) $F(x, y, z) = e^x \sin y\mathbf{i} + e^x \cos y\mathbf{j} + z\mathbf{k}$

(6) Prove that for a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ we have $\text{curl}(\nabla f) = 0$.

(7) Show that for a vector field $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ we have $\text{div}(\text{curl}(F)) = 0$.

(8) Determine whether the vector field $F(x, y, z) = 2xy\mathbf{i} + (x^2 + 2yz)\mathbf{j} + y^2\mathbf{k}$ is conservative and, if so, find a potential function.
